

Takeoff Segment II Distance is the horizontal distance necessary to climb with V_Y from 200 ft to 1000 ft AGL.

Takeoff safety speed is the airspeed that will assure the required climb performance with one engine inoperative.

$$V_{TOSS} = 50 \text{ KIAS}$$

Best rate-of-climb speed is the airspeed which leads to the greatest gain in altitude in a specified time span.

$$V_Y = 60 \text{ KIAS}$$

Wind speed. Charts and calculating examples presented in this supplement do not consider wind speed factors. Apply factors as required by operational rules.

2. LIMITATIONS

2.1 ROTOR RPM LIMITATIONS

EFFECTIVITY *Variants CB and CB-4*

	Power ON	Power OFF
Minimum Transient	85%	—
Minimum Continuous	98%	85%
Maximum Continuous	102%	110%
Maximum Transient (15-sec limited)	105%	—

EFFECTIVITY *Variants CBS and CBS-4*

	Power ON	Power OFF
Minimum Transient	85%	—
Minimum Continuous	98%	85%
Maximum Continuous	102%	104%
Maximum Transient (15-sec limited)	105%	110%

EFFECTIVITY *ALL*

2.2 ENGINE AND TRANSMISSION POWER LIMITATIONS

Engines: Allison 250 - C 20 B

Transmission: Zahnradfabrik Friedrichshafen AG (ZF) FS 72E

WARNING IF TRANSIENT TORQUE/TOT VALUES ARE NOT OBSERVED DUE TO WORK LOAD THEN IT MUST BE ASSUMED THAT THE MAX TORQUE/TOT VALUES HAVE BEEN REACHED AND ONE USAGE OF THE TRANSIENT TORQUE/TOT SHALL BE RECORDED IN THE LOGBOOK.

- CAUTION**
- OEI LIMITS ARE RESTRICTED TO EMERGENCY USE ONLY.
 - FOR POWER CHECK THE ONE ENGINE INOPERATIVE TRANSMISSION LIMIT IS MAX 95% TORQUE.
 - FOR OEI TRAINING PURPOSES SEE PARAGRAPH 9 THIS SUPPLEMENT SUBSECTION.

CONDITION	TRANSMISSION (HELICOPTER) LIMITS MAX TORQUE %	ENGINE OPERATING LIMITS		
		MAX TOT °C	MAX N ₁ %	MAX N ₂ %
Start/shutdown Transient (Max 10 sec above 810 °C)		927		
Transient (Max 6 sec above 810 °C)		843		
Transient (Max 12 sec above 810 °C)		899 ¹⁾		
Transient (Max 15 sec)			106	105
All Engines Operating				
Max Continuous Power	2 x 86	779	105	102
Takeoff Power	2 x 86	810 ²⁾	105	102
One Engine Inoperative				
Max Continuous Power	105	779	105	102
30-min Power	105	810	105	102
2.5-min Power	110	810	105	102
Transient (Max 16 sec above 110%)	123 ³⁾			
NOTE ¹⁾ Max three occurrences per life of each turbine wheel. Logbook entry required. ²⁾ Takeoff Power TOT range of 779-810 °C has a 5-min limit. ³⁾ After a max of ten occurrences above 110% maintenance action is required (see MM). Logbook entry required.				

2.3 WIND LIMITS

Takeoff and landing with a tailwind is prohibited.

2.4 APPROVED LUBRICANTS

Main rotor transmission *) Restricted to EXXON T.O. 25 and BP Turbo Oil 25

*) To be placarded or marked on the equipment.

2.5 INSTRUMENT MARKINGS

The pointers and scales of the instruments listed below are marked as follows:

Start limits	red triangle
Transient limits	red dot
Minimum and maximum limits	red radial
OEI 2.5-min Power	red dashed radial
OEI 30-min Power	yellow dashed radial
Takeoff power or caution range	yellow arc
Normal or continuous range	green arc

2.5.1 Turbine Outlet Temperature (TOT) Indicator

0 to 779 °C	green arc
779 to 810 °C	yellow arc
810 °C	red radial
810 °C	yellow dashed radial
843 °C	red dot
899 °C	red dot
927 °C	red triangle

2.5.2 Dual Torque Indicator

0 to 86%	green arc
86%	red radial
105%	yellow dashed radial
110%	red dashed radial
123%	red dot

2.6 OPTIONAL EQUIPMENT INSTALLATION LIMITATIONS

If an Air Cleaner System (FMS 10-37) is installed, FMS 11-4 operation is only allowed for Air Cleaner Systems with Bleed Air Valve. During FMS 11-4 operation the bypass doors must be open.